The Adelic Physics Programme: Open Problems and Future Directions

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Abstract

The Adelic Physics Programme has produced six published papers spanning the notation problem, the fine-structure constant as a bifurcation parameter, Poisson summation as the adelic bridge, the continuum critique trilogy, FACTORING as representation-dependent complexity, and a falsifiability protocol for the $\mathbb{Q}$-fundamental hypothesis. This paper concludes the programme with a structured research agenda for the remaining open problems and future directions. Five gaps are identified: (G-01) proving the six category-theoretic scaffold-stripping expressions; (G-03) deriving the electron mass scale and muon/tau quantisation from helical topology; (G-07) computing the numerical stability of the discrete null helix; (C-02) verifying the p-adic QEC classifier conjectures; and (L-04) constructing an Adelic Shannon Theory. For each gap, we provide a methodology sketch, effort estimate, success criteria, falsifiability conditions, and connections to existing QNFO infrastructure. The primary executable deliverable is the **G-07 null-helix numerical stability analysis** using discrete differential geometry on GPU — the most tractable high-impact item. The paper concludes with reflections on the programme's achievements, limitations, and the boundary between settled science and open frontier.

Keywords: adelic physics, open problems, future directions, research programme, numerical analysis, discrete differential geometry, helical null-curve, category theory, Shannon theory, quantum error correction

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1. Programme Summary

1.1 What Was Achieved

The Adelic Physics Programme, built on 12 preparatory research notes (2026-07-09 through 2026-07-29) and consolidated in the adelic-epistemological-foundations project, produced six published papers totalling 86 pages:

Paper	Title	DOI	Pages
P1	The Notation Problem: Scaffold-Stripping + DCN	10.5281/zenodo.21691040	13
P2	Alpha as Bifurcation Parameter	10.5281/zenodo.21691059	13
P3	Poisson Summation as the Adelic Bridge	10.5281/zenodo.21691078	12
P4	Continuum Critique Trilogy	10.5281/zenodo.21691415	11
P5	FACTORING + Adelic Complexity	10.5281/zenodo.21691642	13
P6	$\mathbb{Q}$-Fundamental Falsifiability Protocol	10.5281/zenodo.21697717	9

The unifying meta-principle, verified by cross-domain consilience across Physics, Computer Science, Cognitive Science, Information Theory, Biology, and Sociology [1], is: **the real-number continuum is a flat projection that conceals qualitative phase boundaries. Many "unsolved problems" are artifacts of representational choices — continuum, decimal base, container-based notation — rather than genuine ignorance.**

1.2 What Remains

This paper does not repeat what was achieved. It surveys what was NOT achieved — the gaps, open problems, and untested predictions that constitute the programme's future.

The gaps fall into three categories by effort:

Category	Effort	Gaps
Immediate demonstrator	3 months	G-07: α numerical stability analysis
Medium programme	1-2 years	C-02: p-adic QEC classifier verification, L-04: Adelic Shannon Theory
Long-range research	2+ years	G-01: CT expression proofs, G-03: m_e scale and lepton quantisation

Each gap is addressed in a section below.

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2. G-01: Prove the Six Category-Theoretic Expressions

2.1 The Problem

Paper P1 [2] proposed the scaffold-stripping hypothesis: six marginalised formalisms — Laws of Form, Existential Graphs, Viable System Model, Catastrophe Theory, Pattern Language, and Polycontextural Logic — contain valid category-theoretic invariants imprisoned in inaccessible notation. The specific expressions were stated as claims, not proofs [SPECULATIVE]:

Formalism	Claimed Invariant
Laws of Form	Idempotent monad on a 2-category of distinctions
Existential Graphs	Topos-theoretic subobject classifier morphism
Viable System Model	Endofunctor on Sys with fixed point
Catastrophe Theory	Sheaf of singularity unfoldings on stratified space
Pattern Language	Coalgebra for a pattern-composition functor
Polycontextural Logic	Presheaf of Heyting algebras over a contexture site

2.2 Required Work

Proving these expressions requires constructing the mathematical objects:

1. **2-category of distinctions:** Define objects (contexts), 1-morphisms (acts of distinction), and 2-morphisms (transformations of distinctions). Prove that Spencer-Brown's Calling and Crossing rules correspond to the monad axioms (idempotence, $T^2 \cong T$). Show that re-entry corresponds to a fixed point of the monad.

2. **Contexture site:** Define the site whose coverages correspond to Gurwitsch's contextures. Show that the presheaf of truth-values is a sheaf of Heyting algebras, and that Gunther's junctions correspond to gluing conditions.

3. **Stratified space for Catastrophe Theory:** Construct the stratification of the parameter space such that the sheaf of singularity unfoldings is a constructible sheaf. Prove that Thom's seven elementary catastrophes correspond to the possible stalks.

4. **Pattern-composition functor:** Define the endofunctor F on the category of patterns whose coalgebra generates the pattern language. Prove that the terminal coalgebra is the set of all admissible pattern compositions.

2.3 Effort Estimate

Subproblem	Effort	Prerequisites	Difficulty
LoF monad	6 months	2-category theory	Moderate
EG topos	4 months	Topos theory	Moderate
VSM fixed point	3 months	Category theory, cybernetics	Low
CT sheaf	6 months	Singularity theory, sheaf theory	High
PL coalgebra	4 months	Coalgebraic semantics	Moderate
PC presheaf	6 months	Site theory, many-valued logic	High
Total	**~2 years (sequential) or ~1 year (parallel)**	—	—

2.4 Falsifiability

[UNTESTED] The scaffold-stripping hypothesis is falsified if ANY formalism's invariant cannot be expressed in category-theoretic language without loss. A good-faith attempt at constructing the claimed objects that results in a demonstrable failure — e.g., "the 2-category of distinctions does not satisfy the monad axioms because Crossing is not idempotent" — would constitute falsification for that formalism.

2.5 Connection to Existing Infrastructure

The QNFO Quantum Laws of Form paper [3] provides the Syntactic Token Calculus (STC) — a formal system built from the mark and enclosure primitives. The STC can serve as the concrete starting point for constructing the 2-category of distinctions. The STC's reduction rules (Calling, Crossing) provide the equational theory that must be preserved by the monad's multiplication and unit.

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3. G-03: The Electron Mass Scale and Muon/Tau Quantisation

3.1 The Problem

Paper P2 [4] posed three unanswered questions about the helical electron model:

1. **Why m_e $\approx$ 4.2 $\times$ 10^{-23} m_P?** The Compton wavelength λ_C = h/(m_e c) sets the absolute scale of the helical null-curve. In natural units, the electron mass is tiny relative to the Planck mass. The helical model explains the *ratio* r_e/λ_C = α but not the *absolute scale* λ_C.
2. **Why $\alpha \approx$ 1/137?** The stability analysis that would derive α as an eigenvalue of the curvature-torsion energy functional has not been performed (see G-07 below).
3. **Why m_μ/m_e $\approx$ 207 and m_τ/m_e $\approx$ 3477?** The mass ratios of the charged leptons are conjectured to encode topological winding numbers of the helical structure — the electron as n=1, muon as n=207, tau as n=3477 — but this is entirely speculative [SPECULATIVE].

3.2 Approach

The absolute scale problem may be resolved by the Planck-scale discretisation: the Compton wavelength is the smallest length at which a helical null-curve can form on a Planck-space lattice without collapsing into a black hole. If the Planck length ℓ_P is the minimum spatial separation, then the maximum curvature of any null-curve is κ_max $\sim$ 1/ℓ_P, and the minimum pitch (which determines the mass via m = $\hbar\kappa$/c) is determined by the lattice spacing.

This is a numerical general relativity problem requiring a Planck-scale lattice simulation — a highly speculative endeavour [UNTESTED].

3.3 Effort Estimate

This gap represents the hardest problem in the programme. It may require both experimental guidance (the Falsifiability Protocol's all-three-null constraint from P6 [5]) and new mathematical techniques (p-adic general relativity, Bruhat-Tits tree field theory). No timeline estimate is meaningful at this stage.

3.4 Falsifiability

If a fourth charged lepton is discovered with a mass ratio that is NOT a close approximation to an integer (in units of m_e, after accounting for QED radiative corrections), the winding-number conjecture is falsified for the muon and tau as well [STRONG]. As of 2026, no fourth charged lepton has been observed. The LEP collider excluded charged leptons with masses up to ~100 GeV.

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4. G-07: Numerical Stability Analysis of the Discrete Null Helix

4.1 The Problem

Paper P2 [4] posed the helical electron stability problem but did not solve it. The variational problem is:

> Find stationary points of E[γ] = $\int$ ($\kappa^2 - \tau^2$) ds for a null curve γ in Minkowski space, subject to periodicity, non-radiation, and single-valuedness constraints.

This problem is **mathematically open** — no analytic solution is known. However, a **numerical approach** using discrete differential geometry (DDG) on a GPU is computationally feasible and is the most tractable way to make progress.

4.2 Methodology

Step 1: Discretisation (4 weeks). Approximate the null helix as a polygonal chain of N lightlike segments. Each segment is a pair of events (t_i, x_i, y_i, z_i) with null separation. At each vertex, compute:

- **Discrete curvature** κ_i from the turning angle between adjacent segments
- **Discrete torsion** τ_i from the dihedral angle between consecutive osculating planes

The discrete functional is E_discrete = Σ_i (κ_i^2 − τ_i^2) Δs_i.

Step 2: Constraint enforcement (4 weeks). Implement constraints:

- **Periodicity:** $\gamma(0) = \gamma(P)$ for the helical period P
- **Non-radiation:** Compute the Liénard-Wiechert potential for each segment and check that the Poynting flux through a distant sphere is zero (in practice, below a numerical threshold)
- **Single-valuedness:** Ensure the wavefunction phase advance along the curve equals an integer multiple of 2π

Step 3: Optimisation (6 weeks). Minimise E_discrete as a function of the pitch angle θ using:

- Gradient descent with analytic gradient from the DDG expressions
- ADAM optimiser with learning rate annealing
- Stochastic parallel perturbation for escaping local minima
- Population-based sweep (θ from 0 to $\pi/2$ in 10^4 steps)

Step 4: Verification (2 weeks). For the found stationary point θ*:

- Compute the Hessian of E_discrete to confirm it's a minimum (all eigenvalues > 0)
- Check stability against small perturbations of the control points
- Verify that the result is robust to discretisation (vary N from 10^2 to 10^6)

Total: 16 weeks.

4.3 Resource Requirements

Resource	Specification	Estimated Cost
GPU	NVIDIA A100 (80 GB) or better	~$40K or cloud ~$5/hr
DDG library	Geometry Central, libigl, or custom	Open source
Developer time	1 physicist $\times$ 4 months	~$60K (postdoc)
Storage	~500 GB for parameter sweeps	Negligible

Total estimated cost: ~$80–$100K. This is within reach of a standard postdoc/grant budget.

4.4 Success Criteria

The numerical analysis is **successful** if:

1. E_discrete(θ) has a unique minimum at some $\theta^* \in (0, \pi/2)$
2. The Hessian at θ^* is positive definite
3. θ^* is robust to N ($10^2 \leq N \leq 10^6$)
4. θ^* corresponds to $\alpha = \tan(\theta^*) \approx 1/137$ to within numerical precision

The analysis is **informative but inconclusive** if:

1. E_discrete(θ) has multiple minima — α may not be uniquely determined
2. θ^* is discretisation-dependent — the continuum limit is subtle
3. No minimum exists — the helical model is falsified for the chosen functional

4.5 Connection to Existing Infrastructure

The QNFO Ultrametric Engine [6] provides 27+ API endpoints for Bruhat-Tits tree construction and spectral analysis. The DDG framework would extend the existing infrastructure with a new endpoint: `/helical-stability` computing the pitch-angle minimisation for a given N and tolerance.

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5. C-02: p-Adic QEC Classifier Verification

5.1 The Problem

The Number-Theoretic Ultrametric Foundations paper [7] proposed three conjectures:

- **C2.1':** CSS-Ultrametric Correspondence — every CSS code corresponds to an ultrametric tree
- **C5.1:** Kodaira-Néron Fiber Classification for stabiliser codes — code types (CSS, GF(4), stabiliser, graph) correspond to Kodaira-Néron fibre types
- **C7.3':** Mahler v_p-Spectral Decomposition — the p-adic Mahler spectrum distinguishes optimal codes from random ensembles

These conjectures were computationally verified on 4 code families with 83% accuracy [7]. Full verification requires:

1. Expansion to all known QEC codes (surface, topological, colour, LDPC, concatenated, subsystem)
2. Verification of the conjectures on ≥ 100 code families
3. Proof that the classification accuracy generalises beyond the 4 families tested

5.2 Approach

The existing computational engine [6] provides the /spectral-analysis and /validate endpoints. Extending these to the full QEC landscape requires:

- Compiling a comprehensive code database from the literature (~10^3 codes across 10+ families)
- Computing the Mahler v_p spectrum for each code
- Classifying each code's fibre type per C5.1
- Reporting accuracy statistics

5.3 Effort Estimate

Subproblem	Effort	Dependencies
Code database compilation	1 month	QEC literature survey
Mahler spectrum computation	2 months	Existing /spectral-analysis API
Fibre type classification	1 month	Automatic via /validate API
Statistical analysis	1 month	Bootstrapping, cross-validation
Total	**~5 months**	—

Total estimated cost: ~$40K (cloud compute + developer time).

5.4 Falsifiability

[UNTESTED] The conjectures are falsified if classification accuracy on a held-out test set (50 code families, stratified across all known code types) falls below 60%. Random guessing would achieve approximately 25% accuracy for 4 fibre types. A result below 60% would indicate that the ultrametric classification captures signal but not enough to be practically useful.

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6. L-04: Adelic Shannon Theory

6.1 The Problem

The cross-domain consilience audit [1] identified an untested implication in Information Theory:

> If the physical base field is $\mathbb{Q}$ (not $\mathbb{R}$), then Shannon's continuous channel model is an Archimedean approximation to a fundamentally discrete ultrametric channel. p-adic information theory would replace Gaussian noise with ultrametric noise. The "channel capacity" in an adelic communication system would be a product over all places (Archimedean + p-adic), with the Poisson summation formula providing the analytic glue.

The construction of an **Adelic Shannon Theory** — a generalisation of information theory to the adele ring A_$\mathbb{Q}$ — is the most ambitious theoretical gap in the programme [SPECULATIVE].

6.2 Key Components

1. **p-adic entropy:** Define the entropy H__p(X) of a random variable X taking values in ℚ__p, using the p-adic valuation as the cost function rather than the standard logarithm.
2. **Adelic channel capacity:** For a channel whose input and output are adeles, define the capacity C(A__ℚ) = $\prod p\ \ C\_p\ \times\ C\infty$, the product of p-adic and Archimedean capacities. Prove a product-formula coding theorem.
3. **Ultrametric noise model:** Replace the additive white Gaussian noise (AWGN) channel with an additive ultrametric (AUM) channel where noise is p-adic. Show that the AUM capacity is determined by the p-adic valuation of the noise power, not its standard deviation.
4. **Poisson summation as source coding theorem:** Show that the Poisson summation formula IS the data-processing inequality for the adelic source — the Gaussian e^{-π x^2} is the unique source distribution that is invariant under the adelic Fourier transform, analogous to the Gaussian being the maximum-entropy distribution in standard information theory.

6.3 Effort Estimate

Subproblem	Effort	Difficulty
p-adic entropy definition	2 months	Moderate (mathematical)
Adelic channel capacity	6 months	High
Ultrametric noise model	4 months	Moderate
Poisson sum as source coding	6 months	High (requires deep Tate thesis understanding)
Total	**~18 months**	—

6.4 Connections

Adelic Shannon Theory would connect to:

- **P3 [8]:** Poisson summation as the analytic bridge — the mathematical foundation for the source coding theorem
- **P6 [5]:** The falsifiability protocol's ultrametric clustering signature could be reinterpreted as an information-theoretic signature of adelic structure
- **Silent-Radix Cryptography [9]:** A cryptographic instance of place-dependent information content — the same adelic Channel capacity perspective reveals why the silent-radix ambiguity is a genuine information resource, not a limitation

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7. G-07: Detailed Implementation Plan

(The following section serves as the primary executable deliverable for the immediate next phase.)

7.1 Phase Plan

Phase	Timeline	Deliverable	Gate
Phase 1: DDG Library Setup	Weeks 1-2	Geometry Central integration with QNFO/Ultrametric Engine	Compilation, unit tests pass
Phase 2: Null-Curve Discretisation	Weeks 3-4	Polygonal null-helix discretisation of N=10^2 segments	Discrete curvature = continuous curvature to 1%
Phase 3: Constraint Implementation	Weeks 5-8	Periodicity, non-radiation, single-valuedness constraints	All constraints satisfied at N=10^4
Phase 4: Optimisation	Weeks 9-14	Pitch-angle minimisation over $\theta \in (0, \pi/2)$ in 10^4 steps	Stationary point found, Hessian positive-definite
Phase 5: Verification	Weeks 15-16	Robustness checks, discretisation scaling, perturbation analysis	All success criteria from §4.4 satisfied

7.2 Hardware Budget

Item	Weeks	Hours	GPU-hours	Estimated Cost
DDG development	2	160	—	$5K (developer time)
Constraint test (small N)	2	80	100	$1K\|\|Fulloptimisation(N =$ 10^{6}$)
Verification	2	80	500	$2.5K
Total	**16**	**480**	**2,600**	**~$18.5K**

7.3 Success Probability

Subproblem	Probability	Risk	Mitigation
DDG integration	0.90	API incompatibility	Fallback: custom DDG in Python/NumPy
Discrete curvature convergence	0.85	Poor convergence rate	Use higher-order discretisation
Constraint satisfaction	0.70	Non-radiation constraint is computationally expensive	Approximate: check Poynting flux to 1% tolerance

Subproblem	Probability	Risk	Mitigation
Unique minimum found	0.40	Multiple minima or no minimum	This IS the scientific result — publish regardless of finding
Hessian positive-definite	0.50	Stationary point may be saddle	Publish saddle as informative finding, not failure
$\alpha \approx 1/137$	**0.05**	No known derivation has succeeded	This is the most speculative component; a null result is expected

7.4 Publication Plan

Outcome	Paper Type	Target Journal	Probability
$\alpha = 1/137 \pm \epsilon$ confirmed	Letter or Rapid Communication	*Physical Review Letters*	0.05
Stationary point found, not at 1/137	Research article	*Physical Review A* or *Journal of Mathematical Physics*	0.40
No stationary point found	Research article	*Journal of Physics A: Mathematical and Theoretical*	0.30
Methods paper (null-helix DDG framework)	Methods paper	*Computer Physics Communications*	0.25

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8. Programme Conclusion

8.1 What Was Built

The Adelic Physics Programme has produced:

- **6 published papers** (86 pages, 41 references) spanning mathematics (P1, P3), physics (P2), computational complexity (P5), experimental protocol (P6), and unified synthesis (P4)
- **3 red-team audits** with complete kaizen protocol, closing all identified findings
- **1 computational script** verifying the Poisson summation bridge
- **6 durable memories** seeded in Vectorize + D1 + KG
- **15 GitHub commits** on QNFO/adelic-epistemological-foundations

8.2 Limitations

1. **No theorem is proved.** Every paper in the programme is a proposal, analysis, or reframing — not a theorem-proof construction. The scaffold-stripping expressions [2] are claimed, not

proven. The variational problem [4] is posed, not solved. The adelic complexity framework [10] is sketched, not formalised.

2. **No experiment is performed.** The falsifiability protocol [5] identifies three candidate signatures — CMB log-periodic oscillations, ultrametric clustering, rational α fingerprints — but does not perform any of these measurements. The entire programme is currently analytical.
3. **No critical test has been passed or failed.** No experimental result confirms or contradicts the $\mathbb{Q}$-fundamental hypothesis. The programme is in a pre-empirical state.

8.3 The Boundary

The boundary between settled science and open frontier in this programme is:

Settled	Open
The Poisson summation formula is a theorem	Whether it implies $\mathbb{Q}$ is the physical base field is speculation
α is a projective invariant of two electron length scales	Whether it is a geometric eigenvalue is untested
Six formalisms can be expressed in category-theoretic language (claimed)	Whether the expressions are correct is unproved
FACTORING $\in$ BQP is a theorem	FACTORING $\notin$ BPP is unsolved
CMB, ultrametric clustering, and α precision experiments are feasible	Whether any would detect $\mathbb{Q}$-fundamental signatures is unknown

8.4 Invitation

The programme's value lies not in the answers it provides but in the questions it reframes. To the mathematician: can you construct the 2-category of distinctions? To the computational physicist: can you simulate the discrete null helix on GPU? To the experimenter: can you search for log-periodic oscillations in Planck data? To the information theorist: what is the capacity of an adelic channel?

The questions are posed. The tools are specified. The falsifiability conditions are stated. The invitation is open.

Declarations

Funding: This research received no specific grant from any funding agency.

Conflicts of Interest: The author declares no conflicts of interest.

Ethics Approval: Not applicable.

Author Contributions: Single author — all contributions.

Data Availability: No experimental data were generated or analysed.

Code Availability: The DDG framework described in §7 will be released under an open-source license upon completion.

Use of Artificial Intelligence: AI-assisted drafting was used for literature synthesis and prose refinement. All arguments and conclusions were verified by the human author.

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